AP BIOLOGY TOPIC 8.6

Biodiversity, Ecosystem Resilience & Keystone Species

UNIT 8: ECOLOGY (10–15% EXAM WEIGHT)

SYI-3.F | SYI-3.G | KEYSTONE SPECIES | ECOSYSTEM RESILIENCE

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 8.6 (BIODIVERSITY, ECOSYSTEM RESILIENCE & KEYSTONE SPECIES)

CED Core Vocabulary

Three Levels of Biodiversity — 1. Genetic Diversity: Allelic variation within and between populations of a species (enables evolutionary adaptation).

2. Species Diversity — Variety and relative abundance of species within a community (richness + evenness).

3. Ecosystem Diversity — Variety of habitats, biotic communities, and ecological processes across the biosphere.

Short-Term Effects — Immediate population spikes or crashes, altered foraging behavior, trophic cascades, and disrupted nutrient/energy flow.

Long-Term Effects — Permanent loss of native species (reduced biodiversity), evolutionary selection shifts, community restructuring into an alternative state (e.g., kelp forest →...

1. Island Size Effect — LARGER islands have higher biodiversity because they offer more resources/diverse habitats and larger populations, resulting in lower extinction rates .

2. Distance from Mainland Effect — CLOSER islands to the mainland have higher biodiversity because organisms can disperse there much more easily, resulting in higher immigration rates .

§2. 1. BIODIVERSITY & ECOSYSTEM STABILITY / RESILIENCE

Functional Redundancy

- **Three Levels of Biodiversity:**
  - **1. Genetic Diversity:** Allelic variation within and between populations of a species (enables evolutionary adaptation).
  - **2. Species Diversity:** Variety and relative abundance of species within a community (richness + evenness).
  - **3. Ecosystem Diversity:** Variety of habitats, biotic communities, and ecological processes across the biosphere.
- **Why Diverse Ecosystems Are More Resilient (Functional Redundancy):**
  - Ecosystems with **more component parts and higher species diversity** are far more resilient to environmental change (drought, disease, storms, climate shifts, human disturbance).
  - Many species perform **overlapping ecological roles** (nitrogen fixation, pollination, primary production, predation).
  - If one species declines due to a species-specific pathogen or stressor, other species with similar niches step in and **compensate to maintain overall ecosystem function and energy flow**!
- **Low Diversity = High Vulnerability (Natural vs. Artificial Ecosystems):**
  - Ecosystems with few component parts (e.g., **agricultural monoculture farms**, Arctic tundra, isolated islands) lack functional backups; losing a single key species can trigger total food-web collapse.

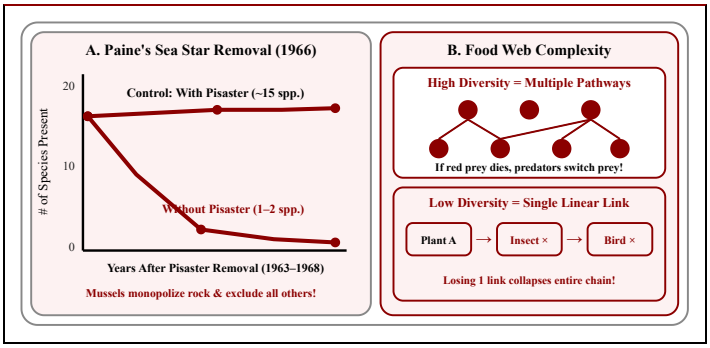
[MECHANISTIC NOTE] **High Biodiversity = Functional Redundancy = High Ecosystem Stability & Resilience.**

[EXAMPLE & BIOLOGICAL APPLICATION] **Natural Tropical Rainforest vs. Monoculture Cornfield:** A diverse rainforest with hundreds of tree species survives a beetle outbreak that kills one tree species, whereas a monoculture crop field can be completely wiped out by a single pest or blight.

[COMMON AP EXAM FRQ TRAP] **Low-diversity systems (like fertilized monoculture farms) may look highly productive in the short term, but they are ecologically unstable and fragile long-term!**

§4. 3. CLASSIC EXPERIMENT: ROBERT PAINE'S PISASTER KEYSTONE STUDY

AP Graph Classic



§5. 4. COMPONENT ADDITION/REMOVAL & ISLAND BIOGEOGRAPHY

Short- vs. Long-Term

- **Adding or Removing a Component from an Ecosystem:**
  - **Short-Term Effects:** Immediate population spikes or crashes, altered foraging behavior, trophic cascades, and disrupted nutrient/energy flow.
  - **Long-Term Effects:** Permanent loss of native species (reduced biodiversity), evolutionary selection shifts, community restructuring into an alternative state (e.g., kelp forest → urchin barren), or total ecosystem collapse.
- **MacArthur & Wilson's Equilibrium Theory of Island Biogeography:**
  - Species richness on an island (or isolated habitat fragment) reaches an equilibrium determined by the balance between **immigration rate of new species** and **extinction rate of established species**:
  - **1. Island Size Effect: LARGER islands** have **higher biodiversity** because they offer more resources/diverse habitats and larger populations, resulting in **lower extinction rates**.
  - **2. Distance from Mainland Effect: CLOSER islands** to the mainland have **higher biodiversity** because organisms can disperse there much more easily, resulting in **higher immigration rates**.
  - *Highest Species Richness = Large Island Close to Mainland!*

§3. 2. KEYSTONE SPECIES, PRODUCERS & ECOSYSTEM ENGINEERS

Disproportionate Impact

ECOLOGICAL ROLE	DEFINITION & ABUNDANCE	CLASSIC AP BIOLOGY EXAMPLES
Keystone Species	Species that exerts a **disproportionately large regulatory effect** on community structure and diversity relative to its **low biomass/abundance**.	• ***Pisaster ochraceus* (Intertidal Sea Star):** Preys on dominant *Mytilus* mussels, preventing competitive exclusion. • **Sea Otters:** Keep sea urchins in check to protect kelp forests. • **Yellowstone Gray Wolves:** Regulate elk populations & browsing behavior, allowing willow/aspen & beaver recovery.
Primary Producers	**High-biomass autotrophs** at the base of the food web that capture solar/chemical energy for all higher trophic levels.	Phytoplankton, grasses, trees, kelp. Without producers, **bottom-up collapse** eliminates all consumers!
Foundation Species / Ecosystem Engineers	Organisms that physically create, modify, or maintain the habitat structure for the entire community.	• **Beavers (Ecosystem Engineer):** Build dams that transform streams into wetland ponds. • **Reef-Building Corals & Kelp (Foundation):** Build physical 3D habitat structure.

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§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 8.6 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 8.6: Biodiversity, Ecosystem Resilience & Keystone Species ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 10, 11 & 12 (Energy Dynamics, Whole-Plant Transpiration & Pillbug Animal Behavior Choice Chambers): Choice chambers testing pillbug (Armadillidium vulgare) chemotaxis/hydrotaxis/kinesis (count pillbugs on wet vs. dry filter paper every 30 s for 10 min; evaluate using χ^2 against 50:50 null distribution). Potometer/gravimetric transpiration across light, wind, and humidity treatments normalized per cm² leaf surface area. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Ecological Equations: Population Growth Rate: $dN/dt = B - D$. Exponential Growth: $dN/dt = r_{max} \cdot N$ (J-curve). Logistic Growth: $dN/dt = r_{max} \cdot N \cdot [(K - N)/K]$ (S-curve; maximum instantaneous growth rate dN/dt occurs at $N = K/2$). Simpson's Diversity Index: $D_I = 1 - \Sigma(n / N)^2$ (ranges from 0 to 1; higher value = higher diversity). Net Primary Productivity: $NPP = GPP - R$. Trophic 10% Rule: ~10% of energy transfers to the next trophic level (~90% lost as metabolic heat/respiration). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Ecological Disruption & Trophic Cascade Case Study: Removal of a keystone predator (e.g., Pisaster ochraceus sea star in rocky intertidal zones, or sea otters in kelp forests) triggers a top-down trophic cascade: prey populations (mussels or sea urchins) explode unchecked, competitively excluding other invertebrates or overgrazing kelp producers, collapsing Simpson's Diversity Index. Similarly, agricultural N/P fertilizer runoff causes Cultural Eutrophication: algal blooms block light, aerobic bacteria decompose dead algae and deplete dissolved O_2, creating hypoxic dead zones.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 8.6 to prove why the claim holds.